

# Towhid Ahmed

Generative AI Engineer · Open to remote — US/EU time zones

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## Professional Summary

Built 4 production-grade LLM systems spanning multi-agent orchestration, advanced RAG, and domain-specific AI applications. Delivered measurable impact: 35% hallucination reduction, 25% relevance improvement, 98% test coverage. Specialized in agentic architectures (LangGraph), enterprise-ready backends (FastAPI), and observable LLMops pipelines (LangSmith). All projects containerized, deployed on cloud, and production-tested.

## Technical Skills

**Generative AI & Agentic Systems:** LangChain, LangGraph, Multi-Agent Systems, Tool Calling, Structured Outputs, Prompt Engineering, RAG (Advanced RAG, Agentic RAG, Multimodal RAG, Corrective RAG, Graph RAG), RAG Pipeline, Retrieval Strategies, Cross-Encoder Re-ranking, Vector Databases (FAISS, ChromaDB, Weaviate)

**LLMops & Deployment:** LangSmith, LLM Guardrails, FastAPI, REST APIs, Docker, Streamlit, CI/CD (GitHub Actions), Render

**Large Language Models & Frameworks:** OpenAI, Groq, LLaMA, Transformers, BERT, Sentence Transformers, Hugging Face

**Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, CNNs, RNNs/LSTMs, Supervised & Unsupervised Learning

**Programming & Tools:** Python, Java, PostgreSQL, SQL, Git, GitHub, Pandas, NumPy

**Software Engineering:** Data Structures & Algorithms (Solved 400+ problems), OOP, LLM System Design

## Technical Experience

**Multi-Agent AI System** — Python, LangGraph, LangChain, FastAPI, Pinecone, LangSmith, Streamlit, Docker, Render, HuggingFace, Tavily GitHub | Demo

- Designed a 4-agent LangGraph workflow (Planner, Retrieval, Search, Synthesizer) routing queries between RAG and web search based on intent and complexity.
- Built a FastAPI backend with async execution and core APIs (/chat, /upload, /health, /metrics) optimized for cloud deployment.
- Developed an end-to-end RAG pipeline with document ingestion, embeddings, Pinecone retrieval, and citation-grounded responses.
- Integrated Tavily with DuckDuckGo fallback and LangSmith tracing for reliable web retrieval and workflow observability.

**Multimodal RAG** — Python, FastAPI, Streamlit, Pinecone, Groq, Llama 3.3, Faster-Whisper, RAG, Vector Search, LangSmith, Docker, Git, GitHub, REST API, Async Programming, Caching GitHub

- Designed a production multimodal retrieval system (PDF/audio/video → BAAI embeddings → Pinecone → cross-encoder re-ranking → Groq LLM) improving relevance by 25% and reducing hallucinations by 35%.
- Implemented two-stage retrieval (semantic chunking + re-ranking) to improve retrieval accuracy.
- Reduced API latency from ~5s to <2s using caching and connection pooling; deployed with Docker and achieved 76-unit test coverage.

**Clinical RAG Assistant** — Python, FastAPI, LangGraph, RAG, Pinecone, LLM Guardrails, Tavily, Streamlit, LangSmith, Docker, JWT GitHub

- Built medical education AI using a 4-agent LangGraph pipeline (Router → Safety → Retrieval → Answer) combining RAG and web search.
- Implemented multi-layer safety system (input filtering, retrieval validation, output sanitization) with JWT-secured APIs.
- Achieved 98/98 passing tests across guardrails, RAG evaluation, and API workflows.

**Advanced RAG System** — Python, LangChain, LLM APIs, Vector Database, Streamlit, RAG GitHub

- Designed an LLM-based retrieval system and AI research assistant for semantic document search and QA over ML research papers
- Built an end-to-end retrieval-augmented generation pipeline covering PDF ingestion, intelligent chunking, embedding generation, vector search, and LLM-based response generation
- Designed a modular architecture separating ingestion, retrieval, LLM orchestration, and UI layers with context-grounded responses to improve answer reliability and reduce hallucinations.

## Education and Certificates

**Education:** Computer Science Engineering(CSE), Chittagong University of Engineering & Technology, Bangladesh

**Certifications:** • Generative AI & Agentic AI — Krish Naik (Hands-on LLM systems & agents)

• LLMops — Udemy (Deployment, monitoring, CI/CD for LLM apps) • Data Structures & Algorithms — TakeUForward